

Twin Counts in Short Intervals: Singular-Series Rigidity and the Density-Gradient Variance*

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Abstract

We measure the variance of the number of twin centres in short intervals on the $6N$ skeleton, the natural Montgomery–Soundararajan question for the twin constellation, sieving all centres to $6N \leq 6 \times 10^8$ (2,166,300 twin centres). Partitioning the centre axis into disjoint windows of length H , the Fano factor $F(H) = \text{Var}/\text{mean}$ traces a clean crossover: it is *sub-Poisson* for small windows (dipping to $F \approx 0.92$ near $H \approx 50$), crosses $F = 1$ at $H^* \approx 410$, and rises steeply super-Poisson thereafter ($F \approx 37$ at $H = 10^5$). We explain the whole curve with two cleanly separable mechanisms. (i) The local pair correlation $C(k)$ of twin centres at separation k is governed exactly by the Hardy–Littlewood singular series of the 4-tuple of offsets $\{-1, 1, 6k-1, 6k+1\}$; we derive the closed form $C(k) = G \prod_{p|k} \frac{p-2}{p-4} \prod_{p|3k \pm 1} \frac{p-3}{p-4}$ with generic constant $G = \prod_{p>3} (1 - \frac{4}{(p-2)^2}) = 0.39688$, and verify it against the data to a median of 1.7% across $k \leq 3000$, reproducing every arithmetic spike. Crucially the integrated local correlation is *negative*, $\sum_k (C(k) - 1) < 0$: the four-fold killer constraint makes twin centres locally *more* regular than Poisson, and this rigidity is exactly the sub-Poisson dip. (ii) The super-Poisson rise is not local at all: it is the smooth deterministic decline of twin density across $[1, X]$. Density-detrending splits the variance into a local part $F_{\text{local}}(H)$, which stays sub-Poisson and tracks the singular-series prediction to $\approx 1\%$, and a trend part $F_{\text{trend}}(H) \propto H$ ($\text{Var}_{\text{trend}} \propto H^2$), which reaches 92% of the total at $H = 10^5$. The crossover H^* marks where the global density gradient overtakes the local arithmetic rigidity. We state plainly what we do *not* resolve: the genuine Montgomery–Soundararajan lower-order term—the $H \log(X/H)$ fluctuation at fixed density and the asymptotic Gaussianity—sits below resolution at reachable H and X , dominated by rigidity and by the deterministic trend; we bound where it must live and leave it open. As throughout, this is a measurement, not a theorem; the Hardy–Littlewood heuristic is taken as input and no claim is made about the infinitude of twins.

1 Introduction

The distribution of primes in short intervals is one of the most-studied lower-order questions in analytic number theory. Writing $\psi(x) = \sum_{n \leq x} \Lambda(n)$, Goldston–Montgomery and then Montgomery–Soundararajan [2] established, conditionally on the Hardy–Littlewood prime-tuple conjecture, that the variance of $\psi(x+H) - \psi(x)$ over $x \in [X, 2X]$ behaves like $H(\log(X/H) - A)$ in a wide range

*Code and measured data: <https://github.com/Ruqing1963/6N-shortinterval-variance>

of H , with an explicit lower-order constant A , and that the fluctuations are asymptotically Gaussian. The leading term is forced by the singular series; the lower-order constant and the limiting distribution are the delicate, genuinely open-feeling content.

This programme has measured the first moment of twin-centre density (Part I), the second-order inter-centre residual (Part IV), the microscopic root $1/(q-2)$ of the mean shift (Part XII), and the ω -conditioned second moment of the prime-factor count (Part XXVI); Part XXVIII audited the twin-enrichment closed form against real centres [10]. What it has *not* done is the spatial short-interval count variance—the Montgomery–Soundararajan question transposed to the twin constellation on the $6N$ skeleton. That is the subject here.

The skeleton makes the experiment clean. Primes > 3 live at $6N \pm 1$; a centre N is a *twin centre* when $6N - 1$ and $6N + 1$ are both prime. We sieve every centre to $6N \leq 6 \times 10^8$ (so $N \leq 10^8$), finding 2,166,300 twin centres at global rate $p_0 = 0.0216630$, partition $[1, N_{\max}]$ into disjoint windows of H consecutive centres, count twin centres per window, and study the Fano factor $F(H) = \text{Var}(\text{count})/\mathbb{E}(\text{count})$ as a function of H . A Poisson process gives $F \equiv 1$; the deviation is the whole signal.

The result is a clean two-mechanism picture. The local arithmetic (the Hardy–Littlewood 4-tuple singular series) makes twin centres *sub*-Poisson at short range—a rigidity we derive in closed form and verify to 1.7%. The smooth decline of twin density across $[1, X]$ makes the count *super*-Poisson at long range. The two cross at $H^* \approx 410$. The genuine Montgomery–Soundararajan lower-order term lives between them, below what $X = 6 \times 10^8$ can resolve; we say so explicitly rather than fit it.

2 Method: two scales in one count

For a window $W = \{a + 1, \dots, a + H\}$ of H consecutive centres, let $T(W) = \sum_{N \in W} \mathbf{1}_{\text{twin}}(N)$. Over the $m = \lfloor N_{\max}/H \rfloor$ disjoint windows we form the sample mean and variance and the Fano factor $F(H)$. Counts are extracted from a single prefix-sum array, so every $F(H)$ is exact for the realised sieve. Writing $\gamma(k) = \mathbb{E}[\mathbf{1}_{\text{twin}}(N)\mathbf{1}_{\text{twin}}(N + k)] - p_0^2$ for the centred pair correlation and $C(k) = 1 + \gamma(k)/p_0^2$ for its normalised form, stationarity gives the exact identity

$$F(H) = (1 - p_0) + \frac{2p_0}{H} \sum_{k=1}^{H-1} (H - k) (C(k) - 1). \quad (1)$$

Equation (1) is the hinge of the paper: it routes the variance entirely through the pair correlation $C(k)$, and $C(k)$ in turn carries two physically distinct contents. At short range $C(k)$ is the *local arithmetic*—the chance that two centres a fixed distance apart are both twin, set by congruence constraints. At long range $C(k)$ also feels the *smooth density gradient*: because the twin density declines over $[1, X]$, even uncorrelated regions contribute a positive offset to $C(k)$ that is nearly independent of k . Separating these two is the analysis.

3 The Fano crossover

Figure 1(A) and Table 1 give the measured $F(H)$. The curve is sub-Poisson for small H , with a shallow minimum $F \approx 0.92$ near $H \approx 30$ –100; it crosses $F = 1$ at $H^* \approx 410$; and it rises steeply afterwards, reaching $F \approx 37$ by $H = 10^5$. A purely Poisson twin process would sit flat at $F = 1$; the data do neither thing a Poisson process would do, and they do them in opposite directions at the two ends. The remainder of the paper identifies the mechanism at each end.

H	F_{total}	F_{local} (detrended)	F_{trend}
10	0.944	0.940	0.003
30	0.924	0.913	0.010
100	0.925	0.890	0.034
300	0.967	0.860	0.102
400	0.997	0.854	0.136
500	1.028	0.850	0.170
1000	1.193	0.836	0.341
3000	1.889	0.819	1.023
10000	4.437	0.868	3.410
30000	11.736	1.030	10.229
100000	36.986	1.322	34.076

Table 1: Measured Fano factor and its decomposition (§5). F_{total} uses all windows; F_{local} is the variance of counts after removing the smooth local density (smoothing scale $W = 10^6$); F_{trend} is the variance of the smooth expected counts. The local part stays sub-Poisson; the trend part grows linearly and dominates at large H . (The F_{local} rise for $H \gtrsim 10^4$ is residual trend leakage as H approaches W , not a local effect; see §6.)

4 The local engine: the Hardy–Littlewood 4-tuple

Two twin centres at N and $N + k$ require the four numbers $6N + \{-1, 1, 6k - 1, 6k + 1\}$ all prime. For a prime $p > 3$ the count of forbidden residues of $N \bmod p$ equals the number of distinct residues $\omega_p(k)$ of the offset set $\{-1, 1, 6k - 1, 6k + 1\} \bmod p$ (multiplication by the unit 6^{-1} and negation are bijections). Generically the four offsets are distinct, $\omega_p = 4$; they collide only when p divides one of the differences $2(3k - 1)$, $6k$, $2(3k + 1)$, i.e. when $p \mid 3k - 1$, $p \mid k$, or $p \mid 3k + 1$ (these events are mutually exclusive for $p > 3$). The local factor of $C(k)$ at p is $(p - \omega_p)p/(p - 2)^2$, giving

$$C(k) = G \prod_{\substack{p \mid k \\ p > 3}} \frac{p - 2}{p - 4} \prod_{\substack{p \mid 3k - 1 \\ p > 3}} \frac{p - 3}{p - 4} \prod_{\substack{p \mid 3k + 1 \\ p > 3}} \frac{p - 3}{p - 4}, \quad G = \prod_{p > 3} \frac{(p - 4)p}{(p - 2)^2} = \prod_{p > 3} \left(1 - \frac{4}{(p - 2)^2}\right). \quad (2)$$

The generic constant evaluates to $G = 0.39688$: at a “neutral” separation (no small prime dividing k or $3k \pm 1$) two twin centres are *anti*-correlated by a factor ≈ 0.40 , the price of the four-fold killer constraint. The corrections are sharp arithmetic boosts: a prime dividing k collapses the two pairs onto $\{-1, 1\} \bmod p$ (factor $(p - 2)/(p - 4)$), and a prime dividing $3k \pm 1$ merges one residue (factor $(p - 3)/(p - 4)$).

Equation (2) is not a fit. Table 2 and Figure 1(B) place it against the measured correlation. The neutral value is recovered at once: $C(1) = 0.4015$ against $G = 0.3969$ ($k = 1$ has no small special prime). Every spike lands: $C(5) = 1.62$ ($5 \mid k$ and $7 \mid 3k - 1$), $C(35) = 2.30$ ($5, 7 \mid k$), $C(210) = 2.25$, $C(2310) = 3.08$. Across all $k \leq 3000$ the median of $|C_{\text{meas}}/C_{\text{HL}} - 1|$ is 1.7%, and the scatter of measured against predicted lies tight on the diagonal (Fig. 2(A)).

The decisive feature is the *sign* of the integrated correlation. Summed over $k \leq 3000$, $\sum (C_{\text{HL}}(k) - 1) = -5.47$: the local arithmetic is net *negative*. Twin centres repel at short range. Through the identity (1) this is exactly the sub-Poisson dip of Figure 1(A): feeding the closed-form C_{HL} into (1) predicts $F \approx 0.89$ at $H = 100$ and $F \approx 0.79$ at $H = 3000$ (Table 3, last column), monotonically sub-Poisson. The local engine, by itself, never produces super-Poisson variance.

k	structure	$C_{\text{meas}}(k)$	$C_{\text{HL}}(k)$ (2)
1	neutral ($\rightarrow G$)	0.4015	0.3969
2	$5 \mid 3k-1, 7 \mid 3k+1$	1.0742	1.0583
5	$5 \mid k, 7 \mid 3k-1$	1.6214	1.5875
7	$7 \mid k, 5 \mid 3k-1$	1.5350	1.5119
35	$5, 7 \mid k$	2.2985	2.2499
77	$7, 11 \mid k$	1.8966	1.8621
210	$5, 7 \mid k$	2.2508	2.2053
2310	$5, 7, 11 \mid k$	3.0814	3.0408

Table 2: The closed form (2) against measured pair correlation at structured separations. Agreement is $\sim 1.5\text{--}2\%$ throughout; the measured values sit uniformly just above the prediction, the signature of the density gradient isolated in §5.

H	F_{meas}	F from C_{meas} (1)	F from C_{HL} (1)
100	0.9254	0.9243	0.8897
300	0.9673	0.9674	0.8597
500	1.0281	1.0281	0.8461
1000	1.1929	1.1907	0.8254
2000	1.5376	1.5336	0.8020
3000	1.8895	1.8853	0.7890

Table 3: The variance identity (1) as cross-check. Fed the *measured* $C(k)$ it reproduces F_{meas} to $\lesssim 0.3\%$ (validating the method). Fed the closed-form $C_{\text{HL}}(k)$ it stays sub-Poisson and *decreasing*—the local arithmetic alone never rises. The gap between the last two columns is the density-gradient content, isolated as F_{trend} in Table 1.

5 The decomposition: rigidity versus density gradient

The measured sum has the opposite sign: $\sum(C_{\text{meas}}(k) - 1) = +45.0$, strongly positive. The reconciliation is visible in Table 2: C_{meas} exceeds C_{HL} by a nearly uniform $\approx 1.5\%$ at *every* k . A small k -independent excess, summed over thousands of separations, swamps the bounded arithmetic deficit and flips the sign of the integral. Such a uniform offset is not arithmetic; it is the smooth decline of twin density across $[1, X]$. Two regions of different density contribute a positive product to the pair count at *all* separations, raising $C(k)$ by the relative variance of the density, here $\approx 1.6\%$ —numerically the observed offset.

We confirm this directly. Removing the smooth local density (a moving average at scale $W = 10^6$, $W \gg H$) from each window count and recomputing the variance gives the detrended Fano F_{local} ; the variance of the smooth expected counts gives F_{trend} . Table 1 and Figure 1(A) show the split. The local part is sub-Poisson throughout the cleanly separated range, dipping to $F_{\text{local}} \approx 0.82$, and it agrees with the singular-series prediction from C_{HL} to about 1%: at $H = 100$, $F_{\text{local}} = 0.890$ against the predicted 0.890; at $H = 300$, 0.860 against 0.860 (Table 3, Fig. 2(B)). The trend part grows linearly, $F_{\text{trend}} \propto H$ (equivalently $\text{Var}_{\text{trend}} \propto H^2$, as a deterministic density ramp must), and overtakes the local part near H^* , reaching 92% of the total variance at $H = 10^5$. The crossover scale $H^* \approx 410$ is precisely where the global density gradient catches up with the local rigidity.

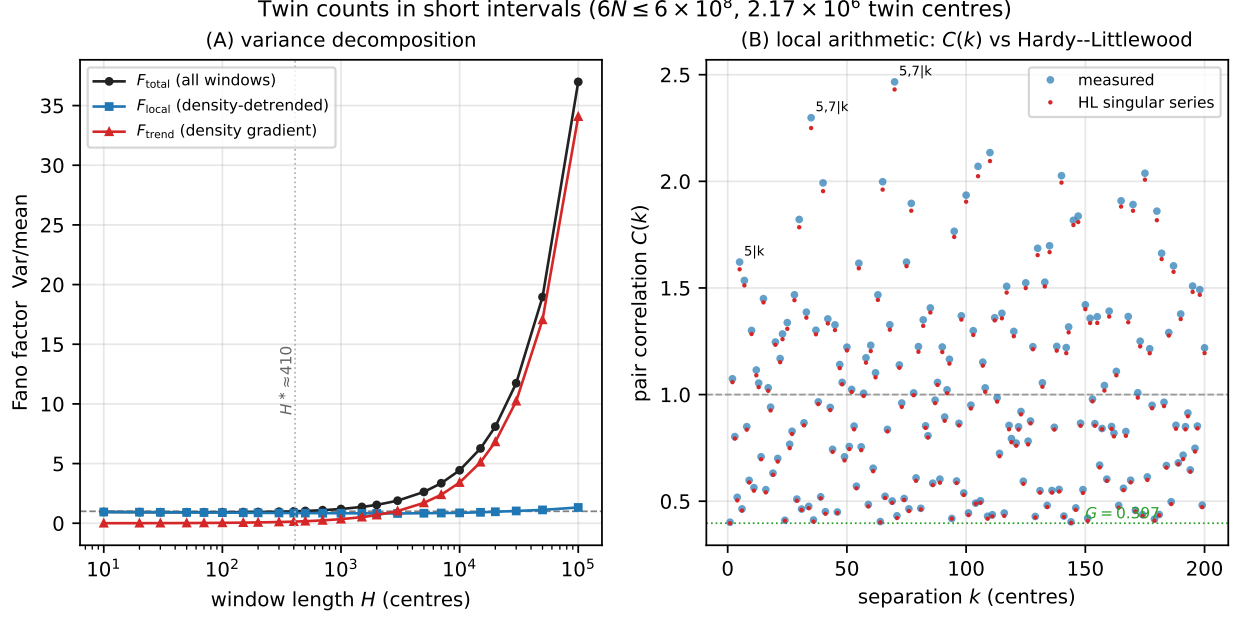


Figure 1: (A) The Fano factor and its decomposition. F_{total} (black) crosses $F = 1$ at $H^* \approx 410$ and rises steeply; the density-detrended local part F_{local} (blue) stays sub-Poisson; the smooth-density-gradient part F_{trend} (red) grows linearly and dominates at large H . (B) The local pair correlation $C(k)$ for $k \leq 200$: measured (blue) against the Hardy–Littlewood closed form (2) (red), with the neutral level $G = 0.397$ marked. Every arithmetic spike is reproduced.

6 Scope and honest ledger

The experiment maps the short-interval twin-count variance completely into two measured mechanisms. The local pair correlation is the Hardy–Littlewood 4-tuple singular series (2), verified to 1.7% in shape and to $\approx 1\%$ in its variance consequence; its integral is negative, so the four-fold killer constraint renders twin centres locally sub-Poisson, and this is the observed dip. The super-Poisson rise is the deterministic decline of twin density across $[1, X]$, with $\text{Var}_{\text{trend}} \propto H^2$, reaching 92% of the variance by $H = 10^5$; the crossover $H^* \approx 410$ separates the two regimes. Both findings are robust: the closed form is parameter free, and the detrended local Fano is stable to the smoothing scale W for $H \ll W$ (varying W from 5×10^5 to 4×10^6 moves F_{local} by $< 1\%$ at $H = 100$ and $< 3\%$ at $H = 3000$).

We are equally clear about what is *not* resolved. The genuine Montgomery–Soundararajan content—the lower-order term in the variance *at fixed density*, the $H \log(X/H)$ fluctuation and its constant, together with the asymptotic Gaussianity of the count [2]—is the quantity our F_{local} approximates. At the reachable H and $X = 6 \times 10^8$ that quantity is *dominated by the singular-series rigidity*: F_{local} *decreases* with H over the clean range, the opposite of the slow logarithmic increase the lower-order term would eventually contribute, so the Montgomery–Soundararajan fluctuation sits below our resolution. We can bound where it must appear—it is hidden beneath a rigidity term of relative size $|\sum(C_{\text{HL}} - 1)|/H$ and beneath a deterministic trend of relative size $\propto H$ —but we cannot isolate its constant or test its Gaussianity at this X . Reaching it would require either a much larger X at fixed octave (to let $\log(X/H)$ grow past the rigidity floor) or a fixed-density dyadic restriction with far more windows than 6×10^8 centres provide. We leave it open, as the

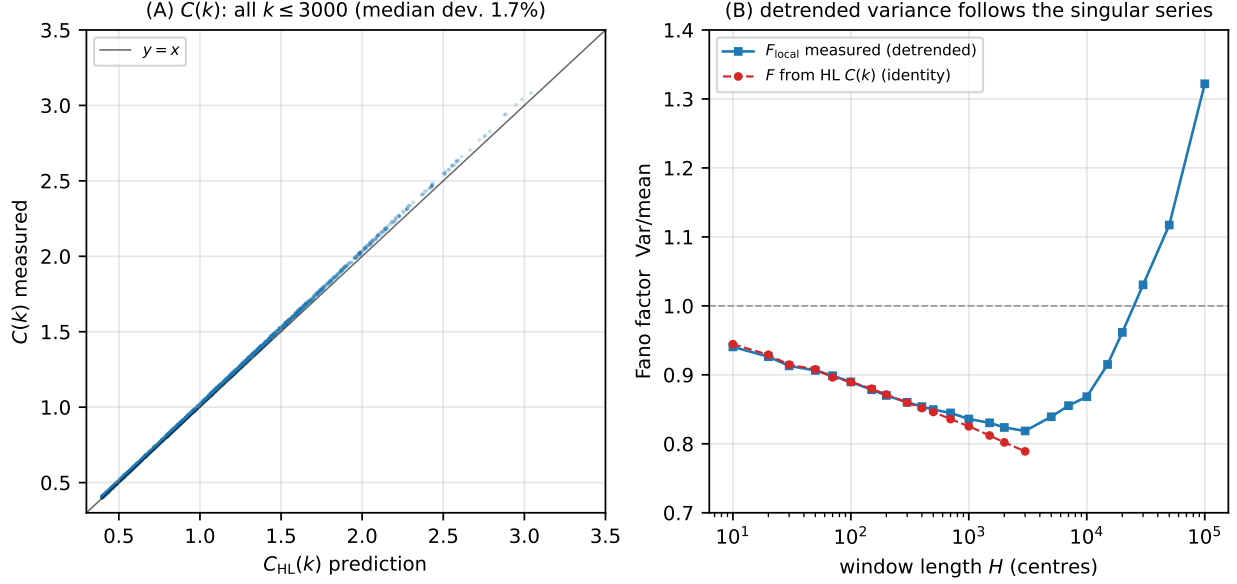


Figure 2: (A) Measured $C(k)$ against the closed form (2) for all $k \leq 3000$; the scatter sits on $y = x$ (median deviation 1.7%) with the slight uniform excess of the density gradient. (B) The density-detrended local Fano F_{local} (blue) follows the prediction obtained by feeding C_{HL} into the variance identity (1) (red), confirming that the local variance is the Hardy–Littlewood singular series and is sub-Poisson.

honest frontier of this question.

Everything here equals the discrete sieve count, takes the Hardy–Littlewood heuristic as input, and makes no claim about the infinitude of twins or of any constellation.

Data and code availability. The sieve (to $6N = 6 \times 10^8$), the Fano-ladder, pair-correlation, Hardy–Littlewood-prediction, variance-identity and density-detrend computations, the robustness scan, and the figure scripts are openly available at <https://github.com/Ruqing1963/6N-shortinterval-variance>.

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